Lower bounds for multivariate independence polynomials and their generalisations

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Abstract

In statistical physics, the multivariate hard-core model describes a system of particles, each of which receives its own fugacity. In graph-theoretic language, the partition function of the model translates to the *multivariate independence polynomial*, i.e., the multiaffine generalisation of the independence polynomial, defined by $Z_G(\lambda_1, \dots, \lambda_n) \coloneqq \sum_{I \in \mathcal{I}(G)} \prod_{v \in I} \lambda_v$, where $\mathcal{I}(G)$ denotes the set of all independent sets in a graph G on $[n] \coloneqq \{1, 2, \dots, n\}$. We prove that for every simple graph G on $[n]$ and $\lambda_1, \dots, \lambda_n \geq 0$,

$$Z_G(\lambda_1, \dots, \lambda_n) \geq \prod_{i=1}^{n} (1 + (d_i + 1)\lambda_i)^{1/(d_i+1)},$$

where $d_1, \dots, d_n$ is the degree sequence of G. This generalises a result of Sah, Sawhney, Stoner, and Zhao, who proved the univariate case $\lambda_1 = \cdots = \lambda_n = \lambda$.

We further conjecture that our inequality should generalise to other antiferromagnetic models and give some evidence in support of it. In particular, for $\lambda_i, \mu_i \geq 0$, $1 \leq i \leq n$, we obtain a stronger inequality

$$\sum_{\substack{I, J \in \mathcal{I}(G) \\ I \cap J = \emptyset}} \prod_{v \in I} \lambda_v \prod_{u \in J} \mu_u \geq \prod_{i=1}^{n} (1 + (d_i + 1)(\lambda_i + \mu_i) + d_i(d_i + 1)\lambda_i \mu_i)^{1/(d_i+1)},$$

which proves our conjecture for a multiaffine generalisation of the *semiproper colouring* partition function with two proper colours.

Our key technical steps for both theorems are obtained by using a custom mathematical research agent built on top of Gemini Deep Think, which can be seen as a benchmark demonstrating that the current state-of-the-art language models can, in part, assist with mathematical research.

1 Introduction

Independence polynomials are a central object in both combinatorics and statistical physics. In particular, its multiaffine generalisation, the *multivariate independence polynomial*, defined by

$$Z_G(\boldsymbol{\lambda}) = \sum_{I \in \mathcal{I}(G)} \prod_{v \in I} \lambda_v,$$

where $\boldsymbol{\lambda} = (\lambda_v)_{v \in V(G)}$ and $\mathcal{I}(G)$ denotes the set of independent sets in G, has played an important role in bridging combinatorics and statistical physics. Indeed, it is the *partition function* of the multivariate hard-core model and furthermore, the *cluster expansion*, which relies on the multivariate Taylor expansion of $\log Z_G(\boldsymbol{\lambda})$, has been an important tool in modern combinatorics; see [DK25; Jen24] for an overview.

With slight abuse of notation, write $Z_G(\lambda) \coloneqq Z_G(\lambda, \dots, \lambda)$, the usual single-variate independence polynomial or, equivalently, the partition function of the hard-core model with fugacity λ. Our first main result is to give a general lower bound for $Z_G(\boldsymbol{\lambda})$ that works for any finite simple graph G and any nonnegative reals λ_v, $v \in V(G)$.

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Theorem 1.1. *Let G be a graph and let $\boldsymbol{\lambda} = (\lambda_v) \in (\mathbb{R}_{\geq 0})^{V(G)}$. Write $d_v \coloneqq \deg_G(v)$. Then*

$$Z_G(\boldsymbol{\lambda}) \geq \prod_{v \in V(G)} Z_{K_{d_v+1}}(\lambda_v)^{\frac{1}{d_v+1}} = \prod_{v \in V(G)} ((d_v+1)\lambda_v + 1)^{\frac{1}{d_v+1}}. \tag{1.1}$$

Equality holds if and only if on each connected component of G, either (i) all $\lambda_v = 0$ or (ii) λ_v is constant and each component of G is a clique.

This generalises a theorem of Sah, Sawhney, Stoner, and Zhao [SSSZ19, Theorem 1.7], who proved (1.1) for single-variate independence polynomials, i.e., the case $\lambda_v = \lambda$ for all $v \in V(G)$. We also remark that the simplest case, when $\lambda_v = 1$ and $d_v = d$ for all $v \in V(G)$, was first proved by Cutler and Radcliffe [CR17, Corollary 2.3]. It is worth noting that, unlike previous results, the inequality (1.1) for $G = K_{d+1}$ is not an equality but an AM–GM-type inequality.

A quick corollary of the inequality (1.1) is a lower bound for the generalised independence polynomial $Z_G(\lambda; \alpha) = \sum_{I \in \mathcal{I}(G)} \lambda^{|I|} \alpha^{e(I,\overline{I})}$, where $e(I, \overline{I})$ is the number of crossing edges between I and $V(G) \setminus I$, a partition function of a 2-spin *antiferromagnetic* model, which we shall define later. Then setting $\lambda_v = \lambda\alpha^{d_v}$ in (1.1) gives the following.

Theorem 1.2. *Let G be a graph and let λ and α be nonnegative reals. Write $d_v \coloneqq \deg_G(v)$. Then*

$$Z_G(\lambda; \alpha) \geq \prod_{v \in V(G)} Z_{K_{d_v+1}}(\lambda; \alpha)^{\frac{1}{d_v+1}} = \prod_{v \in V(G)} \left((d_v+1)\lambda\alpha^{d_v} + 1\right)^{\frac{1}{d_v+1}}.$$

We suspect that this result generalises even further. To elaborate on what this would mean, let H be an edge-weighted graph possibly with loops. Let

$$\hom(G, H) \coloneqq \sum_{\phi: V(G) \to V(H)} \prod_{uv \in E(G)} H(\phi(u), \phi(v))$$

be the (weighted) homomorphism count for G in H, where $H(\cdot, \cdot)$ denotes the corresponding edge weight of H. Then the independence polynomial $Z_G(\lambda)$ for positive rational values $\lambda = q/p$, $p, q \in \mathbb{N}$, corresponds to $\hom(G, H^\lambda)$, where H^λ denotes a suitably 'blown-up' copy of the graph $H =$, i.e., replace the two vertices by p self-looped ones that connect to one another and q non-looped ones, respectively, and put the complete bipartite graph $K_{p,q}$ in between. A graph H with nonnegative edge weights is said to be *antiferromagnetic* if H, as a symmetric matrix with entries corresponding to the edge weights, has exactly one positive eigenvalue, counting multiplicity. For instance, $H =$ is antiferromagnetic, and so are its blow-ups H^λ. We conjecture that any antiferromagnetic H is a 'clique-minimiser' in the following sense.

Conjecture 1.3. *Let G be a graph and write $d_v \coloneqq \deg_G(v)$. Then for any antiferromagnetic edge-weighted graph H,*

$$\hom(G, H) \geq \prod_{v \in V(G)} \hom(K_{d_v+1}, H)^{\frac{1}{d_v+1}}. \tag{1.2}$$

For instance, the Sah–Sawhney–Stoner–Zhao inequality is the aforementioned case H^λ with $H =$. It is not hard to settle the case $H = K_q$, by a slight generalisation of Csikvári's inequality, recorded in Zhao's survey [Zha17, Theorem 8.3]. It can also be seen as an antiferromagnetic counterpart to another theorem of Sah, Sawhney, Stoner, and Zhao [SSSZ20, Theorem 1.14], stating that the reverse inequality to (1.2) holds whenever H is *ferromagnetic*, i.e., all the eigenvalues are nonnegative.

Recall that Theorem 1.2 makes partial progress towards Conjecture 1.3. We make further progress in multiple directions. First of all, we confirm the conjecture for $K_3^\circ \coloneqq$, one loop added to a triangle, in a stronger form that generalises Theorem 1.1. To state the result, for $\boldsymbol{\lambda}, \boldsymbol{\mu} \in \mathbb{R}^{V(G)}$, let

$$Z_G^{(2)}(\boldsymbol{\lambda}, \boldsymbol{\mu}) \coloneqq \sum_{\substack{I, J \in \mathcal{I}(G) \\ I \cap J = \emptyset}} \prod_{v \in I} \lambda_v \prod_{u \in J} \mu_u$$

and write $Z_G^{(2)}(\lambda, \mu) = Z_G^{(2)}(\lambda\mathbf{1}, \mu\mathbf{1})$, where $\mathbf{1} = (1, 1, \ldots, 1)$. In particular, $Z_G^{(2)}(1, 1) = \hom(G, K_3^\circ)$. This can be seen as the number of *semiproper colourings* with two proper colours, i.e., the vertex colouring using two 'proper' colours that cannot make an edge monochromatic and one 'free' colour that can be used with no restrictions. Furthermore, $Z_G^{(2)}(\lambda, \mu)$, $\lambda, \mu > 0$, counts $\hom(G, H)$, where H is obtained by blowing up vertices of K_3° in an appropriate way. For instance, adjusting λ and μ gives a semiproper colouring with two proper colours and multiple free colours too. Our second main result is the following.

Theorem 1.4. *Let G be a graph and let λ_v and μ_v, $v \in V(G)$, be nonnegative reals. Then*

$$Z_G^{(2)}(\boldsymbol{\lambda}, \boldsymbol{\mu}) \geq \prod_{v \in V(G)} Z_{K_{d_v+1}}^{(2)}(\lambda_v, \mu_v)^{\frac{1}{d_v+1}} = \prod_{v \in V(G)} \left((d_v+1)d_v\lambda_v\mu_v + (d_v+1)(\lambda_v+\mu_v)+1\right)^{\frac{1}{d_v+1}}. \quad (1.3)$$

Although letting $\mu_v = 0$ for all $v \in V(G)$ reproves Theorem 1.1, we shall give independent proofs for each of the results.

As an application of our main results, we give slightly weaker lower bounds than (1.2) for semiproper colouring partition functions with q proper colours, $q > 2$, which can be seen as an 'approximate version' of Conjecture 1.3 for semiproper colourings; see Theorem 4.3. Moreover, we also settle the conjecture for the case $\Delta(G) \leq 2$, where $\Delta(G)$ is the maximum degree of G; see Theorem 4.1.

An interesting feature of Theorems 1.1 and 1.4 is that the key steps in both proofs are obtained, at least in part, by using *Aletheia*, a mathematics research agent built upon Gemini Deep Think at Google DeepMind, under the lead of Tony Feng. We will point out these points of contribution where they arise. We also refer to the forthcoming paper [Fen+26] for more details of the development of *Aletheia* and its applications to various problems.

2 Multivariate independence polynomials

The starting point of our proof of Theorem 1.1 is the standard vertex-deletion recurrence for the independence polynomial. Namely, for a graph G and its vertex w, conditioning on whether w is occupied,

$$Z_G(\lambda) = Z_{G\setminus w}(\lambda) + \lambda Z_{G\setminus(\{w\}\cup N(w))}(\lambda),$$

where $N(w)$ is the set of neighbours of w in G. Equivalently, in the multivariate setting, deleting $N(w)$ is the same as setting $\lambda_v = 0$ for each $v \in N(w)$. This recurrence and its variants have been used in several places to bound the number of independent sets [GZ11; SSSZ19; SSSZ20; BHK25], and it also underlies inductive lower bounds on the independence number $\alpha(G)$ (see, e.g., [She83]).

Let us formalise this inductive idea. First, both sides of (1.1) factor over connected components of G, so we may assume that G is connected. For a graph G and $\boldsymbol{\lambda} = (\lambda_v)_{v\in V(G)}$, let

$$Z_G^-(\boldsymbol{\lambda}) := \prod_{v \in V(G)} Z_{K_{d_v+1}}(\lambda_v)^{\frac{1}{d_v+1}}.$$

Then Theorem 1.1 states that $Z_G(\boldsymbol{\lambda}) \geq Z_G^-(\boldsymbol{\lambda})$ for all $\boldsymbol{\lambda} \in (\mathbb{R}_{\geq 0})^{V(G)}$. We use induction on $|V(G)|$ to prove this inequality, where the base case $|V(G)| = 1$ is clear.

Let w be a vertex in G with maximum degree Δ. Fix $\boldsymbol{\lambda} = (\lambda_v)_{v\in V(G)} \in (\mathbb{R}_{\geq 0})^{V(G)}$ and let $\boldsymbol{\lambda}' := (\lambda_v)_{v\in V(G\setminus w)}$. We begin with the multivariate recursive expression

$$Z_G(\boldsymbol{\lambda}) = Z_{G\setminus w}(\boldsymbol{\lambda}') + \lambda_w Z_{G\setminus w}(\boldsymbol{\lambda}')|_{\lambda_v=0,\, v\in N(w)}.$$

By the induction hypothesis, it suffices to show that

$$Z_{G\setminus w}^-(\boldsymbol{\lambda}') + \lambda_w Z_{G\setminus w}^-(\boldsymbol{\lambda}')|_{\lambda_v=0, v\in N(w)} \geq Z_G^-(\boldsymbol{\lambda}). \quad (2.1)$$

The definition gives

$$Z_{G\setminus w}^-(\boldsymbol{\lambda}') = \prod_{v\in N(w)} Z_{K_{d_v}}(\lambda_v)^{\frac{1}{d_v}} \prod_{v\in V(G)\setminus(\{w\}\cup N(w))} Z_{K_{d_v+1}}(\lambda_v)^{\frac{1}{d_v+1}},$$

so after removing the common factor $\prod_{v\in V(G)\setminus(\{w\}\cup N(w))} Z_{K_{d_v+1}}(\lambda_v)^{\frac{1}{d_v+1}}$, (2.1) is equivalent to

$$\prod_{v\in N(w)} Z_{K_{d_v}}(\lambda_v)^{\frac{1}{d_v}} + \lambda_w \geq Z_{K_{\Delta+1}}(\lambda_w)^{\frac{1}{\Delta+1}} \prod_{v\in N(w)} Z_{K_{d_v+1}}(\lambda_v)^{\frac{1}{d_v+1}}, \quad (2.2)$$

which is shown by the following lemma. We remark that this result is exactly what *Aletheia* successfully derived. We integrated this with the correct solutions provided by other models, including Gemini Deep Think 3.0 and its IMO-Gold version, to formulate a cleaner argument.

Lemma 2.1. *Let $\Delta \geq 1$ be an integer and $1 \leq d_1, \dots, d_\Delta \leq \Delta$. Let $\lambda_0, \lambda_1, \dots, \lambda_\Delta$ be nonnegative reals. Then*

$$\prod_{i=1}^{\Delta}(1+d_i\lambda_i)^{\frac{1}{d_i}} + \lambda_0 \geq (1+(\Delta+1)\lambda_0)^{\frac{1}{\Delta+1}} \prod_{i=1}^{\Delta}(1+(d_i+1)\lambda_i)^{\frac{1}{d_i+1}}.$$

Proof. Let $A \coloneqq \prod_{i=1}^{\Delta} A_i$ and $B \coloneqq \prod_{i=1}^{\Delta} B_i$ where

$$A_i \coloneqq (1+d_i\lambda_i)^{\frac{1}{d_i}} \quad \text{and} \quad B_i \coloneqq (1+(d_i+1)\lambda_i)^{\frac{1}{d_i+1}}.$$

The desired inequality then rephrases as

$$A + \lambda_0 \geq (1+(\Delta+1)\lambda_0)^{\frac{1}{\Delta+1}} B. \tag{2.3}$$

By the AM–GM inequality,

$$\begin{aligned}(1+(\Delta+1)\lambda_0)^{\frac{1}{\Delta+1}} B &= \big((1+(\Delta+1)\lambda_0)\cdot(B^{\frac{\Delta+1}{\Delta}})^{\Delta}\big)^{\frac{1}{\Delta+1}} \\ &\leq \frac{(1+(\Delta+1)\lambda_0) + \Delta B^{\frac{\Delta+1}{\Delta}}}{\Delta+1} = \frac{1+\Delta B^{\frac{\Delta+1}{\Delta}}}{\Delta+1} + \lambda_0,\end{aligned}$$

so it suffices to show that

$$(\Delta+1)A \geq \Delta B^{\frac{\Delta+1}{\Delta}} + 1. \tag{2.4}$$

The proof of this inequality consists of two technical steps. Firstly, we prove that, for each $1 \leq i \leq \Delta$,

$$(\Delta+1)A_i^{\Delta} = (\Delta+1)(1+d_i\lambda_i)^{\frac{\Delta}{d_i}} \geq 1 + \Delta(1+(d_i+1)\lambda_i)^{\frac{\Delta+1}{d_i+1}} = 1 + \Delta B_i^{\Delta+1}, \tag{2.5}$$

multiplying which for all $1 \leq i \leq \Delta$ gives

$$(\Delta+1)^{\Delta} A^{\Delta} \geq \prod_{i=1}^{\Delta}(1+\Delta B_i^{\Delta+1}). \tag{2.6}$$

The second step is to show that

$$\prod_{i=1}^{\Delta}(1+\Delta B_i^{\Delta+1})^{\frac{1}{\Delta}} \geq 1 + \Delta B^{\frac{\Delta+1}{\Delta}}. \tag{2.7}$$

Combining this with (2.6) concludes the proof of (2.4).

To prove the first inequality (2.5), the following claim with $d = d_i$ and $\lambda = \lambda_i$ suffices.

Claim 2.2. *For $1 \leq d \leq \Delta$ and $\lambda \geq 0$,*

$$(\Delta+1)(1+d\lambda)^{\Delta/d} \geq 1 + \Delta(1+(d+1)\lambda)^{(\Delta+1)/(d+1)}.$$

Proof. Let

$$f(\lambda) \coloneqq (\Delta+1)(1+d\lambda)^{\Delta/d} - \Delta(1+(d+1)\lambda)^{(\Delta+1)/(d+1)} - 1,$$

then we want to show $f(\lambda) \geq 0$. Since $f(0) = 0$, it suffices to show $f'(\lambda) \geq 0$ for all $\lambda \geq 0$. We have

$$\begin{aligned}f'(\lambda) &= \Delta(\Delta+1)(1+d\lambda)^{\Delta/d-1} - \Delta(\Delta+1)(1+(d+1)\lambda)^{(\Delta+1)/(d+1)-1} \\ &= \Delta(\Delta+1)\big((1+d\lambda)^{(\Delta-d)/d} - (1+(d+1)\lambda)^{(\Delta-d)/(d+1)}\big),\end{aligned}$$

which is nonnegative if

$$(1+d\lambda)^{(d+1)/d} \geq 1 + (d+1)\lambda.$$

This is a consequence of Bernoulli's inequality $(1+x)^{\alpha} \geq 1 + \alpha x$, for $x \geq 0$ and $\alpha \geq 1$. ■

It remains to show (2.7). Let $y_i \coloneqq \Delta B_i^{\Delta+1}$. Then (2.7) is equivalent to

$$\prod_{i=1}^{\Delta}(1+y_i)^{1/\Delta} \geq 1 + \prod_{i=1}^{\Delta} y_i^{1/\Delta}.$$

This follows from the AM–GM inequality of the form

$$1 = \frac{1}{\Delta}\left(\sum_{i=1}^{\Delta}\frac{1}{1+y_i} + \sum_{i=1}^{\Delta}\frac{y_i}{1+y_i}\right) \geq \prod_{i=1}^{\Delta}\frac{1}{(1+y_i)^{1/\Delta}} + \prod_{i=1}^{\Delta}\frac{y_i^{1/\Delta}}{(1+y_i)^{1/\Delta}},$$

Indeed, multiplying $\prod_{i=1}^{\Delta}(1+y_i)^{1/\Delta}$ on both sides concludes the proof. □

Remark. It is not hard to see that equality in Lemma 2.1 holds if and only if either

(i) $\lambda_0 = \lambda_1 = \cdots = \lambda_\Delta = 0$ or

(ii) $\lambda_0 = \lambda_1 = \cdots = \lambda_\Delta \neq 0$ and $d_1 = \cdots = d_\Delta = \Delta$.

Using this, it follows that the equality in (1.1) holds if and only if for each connected component C of G,

(a) $\lambda_v = 0$ for all $v \in C$ or

(b) $\lambda_v = \lambda \neq 0$ for all $v \in C$ and $G[C]$ is a clique.

To see this, we use induction on $|V(G)|$, where the base case is clear. We may assume that G is connected. Suppose the equality in (1.1) holds so that $Z_G(\boldsymbol{\lambda}) = Z_G^-(\boldsymbol{\lambda})$. Recall that $Z_G(\boldsymbol{\lambda}) \geq Z_G^-(\boldsymbol{\lambda})$ follows from

$$\begin{aligned} Z_G(\boldsymbol{\lambda}) &= Z_{G\setminus w}(\boldsymbol{\lambda}') + \lambda_w Z_{G\setminus w}(\boldsymbol{\lambda}')|_{\lambda_v=0,\, v\in N(w)} \\ &\geq Z_{G\setminus w}^-(\boldsymbol{\lambda}') + \lambda_w Z_{G\setminus w}^-(\boldsymbol{\lambda}')|_{\lambda_v=0,\, v\in N(w)} \qquad (2.8) \\ &\geq Z_G^-(\boldsymbol{\lambda}), \qquad (2.9) \end{aligned}$$

and that (2.9) is equivalent to (2.2), where the latter holds by Lemma 2.1. Equality occurs in (2.8) if and only if both of the following hold:

(A1) $Z_{G\setminus w}(\boldsymbol{\lambda}') = Z_{G\setminus w}^-(\boldsymbol{\lambda}')$.

(A2) $\lambda_w = 0$ or $Z_{G\setminus w}^-(\boldsymbol{\lambda}')|_{\lambda_v=0,\, v\in N(w)} = Z_{G\setminus w}(\boldsymbol{\lambda}')|_{\lambda_v=0,\, v\in N(w)}$.

(A1) occurs, by the induction hypothesis, only if λ_v is constant on each connected component of $G\setminus w$. In addition, (A2) occurs, again by the induction hypothesis, if and only if either (B1) $\lambda_w = 0$ or (B2) $\lambda_v = 0$ for all $v \in V(G) \setminus (\{w\} \cup N(w))$.

Equality occurs in (2.9), by the equality case of (2.2), if and only if either

(C1) $\lambda_w = \lambda_v = 0$ for all $v \in N(w)$ or

(C2) $\lambda_w = \lambda_v$ and $d_v = \Delta$ for all $v \in N(w)$.

In either of the cases, since G is connected and λ_v is constant on each connected component of $G \setminus w$, λ_v is constant on $V(G)$. If (C1) occurs, $\lambda_v = 0$ for all $v \in V(G)$. Otherwise, suppose (C2) occurs but $\lambda_w = \lambda_v \neq 0$ for all $v \in N(w)$. Then (B1) is not the case, so (B2) must occur. This implies that $V(G) \setminus (\{w\} \cup N(w))$ is empty, since otherwise $\lambda_v = 0$ for all $v \in V(G)$. Thus, $V(G) = \{w\} \cup N(w)$, and since $d_v = \Delta$ for all $v \in N(w)$, G must be a clique.

3 Semiproper colourings

In our proof of Theorem 1.4, we used *Aletheia* in a more complex and rather interactive way than in the proof of Theorem 1.1, where we essentially outsourced one key lemma. Because of this, we shall first give a full proof and then discuss how the AI agent helped along the way in the remark at the very end of this section.

To prove Theorem 1.4, we may assume that G is connected, analogously to the proof of Theorem 1.1. For a graph G and $\boldsymbol{\lambda} = (\lambda_v)_{v\in V(G)}$, $\boldsymbol{\mu} = (\mu_v)_{v\in V(G)}$, let

$$Z_G^{(2)-}(\boldsymbol{\lambda}, \boldsymbol{\mu}) := \prod_{v\in V(G)} Z_{K_{d_v+1}}^{(2)}(\lambda_v, \mu_v)^{\frac{1}{d_v+1}}.$$

Then Theorem 1.4 states that $Z_G^{(2)}(\boldsymbol{\lambda}, \boldsymbol{\mu}) \geq Z_G^{(2)-}(\boldsymbol{\lambda}, \boldsymbol{\mu})$ for all $\boldsymbol{\lambda}, \boldsymbol{\mu} \in (\mathbb{R}_{\geq 0})^{V(G)}$. To prove this inequality, we again use the idea of 'conditioning on the state assigned to w' to obtain a recurrence.

To elaborate, we proceed by induction on $|V(G)|$, where the base case $|V(G)| = 1$ holds clearly. Let w be a vertex in G with maximum degree Δ. Fix $\boldsymbol{\lambda}, \boldsymbol{\mu} \in (\mathbb{R}_{\geq 0})^{V(G)}$ and let $\boldsymbol{\lambda}' := (\lambda_v)_{v\in V(G\setminus w)}$ and $\boldsymbol{\mu}' := (\mu_v)_{v\in V(G\setminus w)}$. The recurrence relation obtained by fixing the image of w into K_3° is

$$Z_G^{(2)}(\boldsymbol{\lambda}, \boldsymbol{\mu}) = Z_{G\setminus w}^{(2)}(\boldsymbol{\lambda}', \boldsymbol{\mu}') + \lambda_w Z_{G\setminus w}^{(2)}(\boldsymbol{\lambda}', \boldsymbol{\mu}')|_{\lambda_v=0,\, v\in N(w)} + \mu_w Z_{G\setminus w}^{(2)}(\boldsymbol{\lambda}', \boldsymbol{\mu}')|_{\mu_v=0,\, v\in N(w)}.$$

Using the induction hypothesis, it suffices to show that

$$Z^{(2)-}_{G\setminus w}(\boldsymbol{\lambda}',\boldsymbol{\mu}') + \lambda_w Z^{(2)-}_{G\setminus w}(\boldsymbol{\lambda}',\boldsymbol{\mu}')|_{\lambda_v=0,\, v\in N(w)} + \mu_w Z^{(2)-}_{G\setminus w}(\boldsymbol{\lambda}',\boldsymbol{\mu}')|_{\mu_v=0,\, v\in N(w)} \geq Z^{(2)-}_{G}(\boldsymbol{\lambda},\boldsymbol{\mu}). \tag{3.1}$$

By definition,

$$Z^{(2)-}_{G\setminus w}(\boldsymbol{\lambda}',\boldsymbol{\mu}') = \prod_{v\in N(w)} Z^{(2)}_{K_{d_v}}(\lambda_v,\mu_v)^{\frac{1}{d_v}} \prod_{v\in V(G)\setminus(\{w\}\cup N(w))} Z^{(2)}_{K_{d_v+1}}(\lambda_v,\mu_v)^{\frac{1}{d_v+1}},$$

so after removing the common factors $Z^{(2)}_{K_{d_v+1}}(\lambda_v,\mu_v)^{\frac{1}{d_v+1}}$ for $v\in V(G)\setminus(\{w\}\cup N(w))$, (3.1) becomes

$$\prod_{v\in N(w)} A_{d_v}(\lambda_v,\mu_v)^{\frac{1}{d_v}} + \lambda_w \prod_{v\in N(w)} B_{d_v}(\mu_v)^{\frac{1}{d_v}} + \mu_w \prod_{v\in N(w)} B_{d_v}(\lambda_v)^{\frac{1}{d_v}}$$
$$\geq A_{\Delta+1}(\lambda_w,\mu_w)^{\frac{1}{\Delta+1}} \prod_{v\in N(w)} A_{d_v+1}(\lambda_v,\mu_v)^{\frac{1}{d_v+1}}, \tag{3.2}$$

where for $d\geq 1$ and $\lambda,\mu\geq 0$,

$$A_d(\lambda,\mu) := d(d-1)\lambda\mu + d(\lambda+\mu) + 1 = Z^{(2)}_{K_d}(\lambda,\mu) \quad \text{and}$$
$$B_d(\lambda) := d\lambda + 1 = Z^{(2)}_{K_d}(\lambda,0) = Z_{K_d}(\lambda).$$

It is convenient to treat the case $\Delta=1$ separately to assume $\Delta\geq 2$. Suppose $\Delta=1$ and let v be the unique neighbour of w. Since $d_v=\Delta=1$, (3.2) is written as

$$A_{d_v}(\lambda_v,\mu_v) + \lambda_w B_{d_v}(\mu_v) + \mu_w B_{d_v}(\lambda_v) \geq A_2(\lambda_w,\mu_w)^{1/2} A_2(\lambda_v,\mu_v)^{1/2}. \tag{3.3}$$

Let $x_1 := \lambda_w+1$, $y_1 := \mu_w+1$, $x_2 := \lambda_v+1$, and $y_2 := \mu_v+1$. Then (3.3) is equivalent to

$$x_1y_2 + y_1x_2 - 1 \geq (2x_1y_1-1)^{1/2}(2x_2y_2-1)^{1/2}.$$

The AM–GM inequality gives

$$x_1y_2 + y_1x_2 - 1 \geq 2(x_1y_2y_1x_2)^{1/2} - 1.$$

Substituting $x=(x_1y_1)^{1/2}$ and $y=(x_2y_2)^{1/2}$ into $(2xy-1)^2-(2x^2-1)(2y^2-1) = 2(x-y)^2\geq 0$ gives

$$2(x_1y_1x_2y_2)^{1/2} - 1 \geq (2x_1y_1-1)^{1/2}(2x_2y_2-1)^{1/2}$$

and hence, (3.3) holds. Therefore, in what follows, we assume $\Delta\geq 2$.

3.1 The dual set $\mathcal{S}_d$ and log-convexity

For $d\geq 2$, let

$$\mathcal{S}_d := \{(a_0,a_1,a_2)\in(\mathbb{R}_{>0})^3 : a_0 + a_1x + a_2y \geq A_{d+1}(x,y)^{\frac{1}{d+1}} \ \forall x,y\geq 0\}.$$

One good reason to look into this set is that $A_{d+1}(x,y)^{\frac{1}{d+1}}$ is concave, which means $\mathcal{S}_d$ can be seen as a 'dual' of it. Concavity of $A_{d+1}^{\frac{1}{d+1}}$ is straightforward to check.

Lemma 3.1. *For $d\geq 2$, $A_{d+1}(x,y)^{\frac{1}{d+1}}$ is concave on any convex subset of $\{(x,y)\in\mathbb{R}^2 : A_{d+1}(x,y)>0\}$.*

Proof. Let Ω be a convex subset of $\{(x,y)\in\mathbb{R}^2 : A_{d+1}(x,y)>0\}$. Direct calculation gives

$$\partial_{xx}\big(A_{d+1}(x,y)^{\frac{1}{d+1}}\big) = -d(dy+1)^2 A_{d+1}(x,y)^{\frac{1}{d+1}-2} \leq 0, \tag{3.4}$$
$$\partial_{yy}\big(A_{d+1}(x,y)^{\frac{1}{d+1}}\big) = -d(dx+1)^2 A_{d+1}(x,y)^{\frac{1}{d+1}-2} \leq 0, \tag{3.5}$$
$$\partial_{xy}\big(A_{d+1}(x,y)^{\frac{1}{d+1}}\big) = d(dxy+x+y) A_{d+1}(x,y)^{\frac{1}{d+1}-2}.$$

Then the Hessian determinant of $A_{d+1}(x,y)^{\frac{1}{d+1}}$ is

$$d^2 A_{d+1}(x,y)^{\frac{2}{d+1}-3}\big((d-1)(dxy+x+y)+1\big).$$

Since $A_{d+1}(x,y) = (d+1)(dxy + x + y) + 1 > 0$ on Ω,

$$(d-1)(dxy + x + y) + 1 > -\frac{d-1}{d+1} + 1 > 0,$$

so $A_{d+1}(x,y)^{\frac{1}{d+1}}$ has a positive Hessian determinant on Ω. Together with (3.4) and (3.5), we conclude that $A_{d+1}(x,y)^{\frac{1}{d+1}}$ is concave on Ω. □

Now our goal (3.2) is equivalent to the following lemma.

Lemma 3.2. *Let $\Delta \geq 2$ and $1 \leq d_i \leq \Delta$, $1 \leq i \leq \Delta$ be integers. Let $\lambda_i, \mu_i \geq 0$ for $1 \leq i \leq \Delta$. Then*

$$\left(\prod_{i=1}^{\Delta} \frac{A_{d_i}(\lambda_i,\mu_i)^{\frac{1}{d_i}}}{A_{d_i+1}(\lambda_i,\mu_i)^{\frac{1}{d_i+1}}}, \prod_{i=1}^{\Delta} \frac{B_{d_i}(\mu_i)^{\frac{1}{d_i}}}{A_{d_i+1}(\lambda_i,\mu_i)^{\frac{1}{d_i+1}}}, \prod_{i=1}^{\Delta} \frac{B_{d_i}(\lambda_i)^{\frac{1}{d_i}}}{A_{d_i+1}(\lambda_i,\mu_i)^{\frac{1}{d_i+1}}}\right) \in \mathcal{S}_\Delta.$$

Our plan is to show that $\log \mathcal{S}_\Delta := \{(\log a_0, \log a_1, \log a_2) : (a_0,a_1,a_2) \in \mathcal{S}_\Delta\}$ is convex or, equivalently, for all $(a_0,a_1,a_2), (b_0,b_1,b_2) \in \mathcal{S}_\Delta$, $(\sqrt{a_0b_0}, \sqrt{a_1b_1}, \sqrt{a_2b_2}) \in \mathcal{S}_\Delta$; see Lemma 3.4. Then Lemma 3.2 is a consequence of the following lemma with $(d,\lambda,\mu) = (d_i,\lambda_i,\mu_i)$ for all $1 \leq i \leq \Delta$, together with convexity of $\log \mathcal{S}_\Delta$, which allows us to take coordinatewise geometric mean.

Lemma 3.3. *Let $\Delta \geq 2$ and $1 \leq d \leq \Delta$ be integers. Let $\lambda, \mu \geq 0$. Then*

$$\left(\frac{A_d(\lambda,\mu)^{\frac{\Delta}{d}}}{A_{d+1}(\lambda,\mu)^{\frac{\Delta}{d+1}}}, \frac{B_d(\mu)^{\frac{\Delta}{d}}}{A_{d+1}(\lambda,\mu)^{\frac{\Delta}{d+1}}}, \frac{B_d(\lambda)^{\frac{\Delta}{d}}}{A_{d+1}(\lambda,\mu)^{\frac{\Delta}{d+1}}}\right) \in \mathcal{S}_\Delta.$$

To illustrate what Lemma 3.3 means, it is worth noting that the case $d = \Delta$ amounts to calculating the tangent plane of $z = A_{d+1}(x,y)^{\frac{1}{d+1}}$ at $(x,y) = (\lambda,\mu)$. Indeed,

$$\partial_x\left(A_{d+1}^{\frac{1}{d+1}}\right) = \frac{1}{d+1} A_{d+1}^{-\frac{d}{d+1}} \partial_x A_{d+1} = A_{d+1}^{-\frac{d}{d+1}} B_d(y) \quad \text{and} \quad \partial_y\left(A_{d+1}^{\frac{1}{d+1}}\right) = A_{d+1}^{-\frac{d}{d+1}} B_d(x),$$

so the tangent plane at $(x,y) = (\lambda,\mu)$ is

$$\begin{aligned} z &= A_{d+1}(\lambda,\mu)^{-\frac{d}{d+1}}\big(B_d(\mu)(x-\lambda) + B_d(\lambda)(y-\mu)\big) + A_{d+1}(\lambda,\mu)^{\frac{1}{d+1}} \\ &= A_{d+1}(\lambda,\mu)^{-\frac{d}{d+1}}\big(B_d(\mu)(x-\lambda) + B_d(\lambda)(y-\mu) + A_{d+1}(\lambda,\mu)\big) \\ &= A_{d+1}(\lambda,\mu)^{-\frac{d}{d+1}}\big(B_d(\mu)x + B_d(\lambda)y + A_d(\lambda,\mu)\big). \end{aligned} \tag{3.6}$$

As $A_{d+1}(x,y)^{\frac{1}{d+1}}$ is concave, this plane always lies above $z = A_{d+1}(x,y)^{\frac{1}{d+1}}$, which proves the case $d = \Delta$.

Having seen this, Lemma 3.3 essentially states that $(a_0,a_1,a_2) \in \mathcal{S}_d$ implies $(a_0^{\Delta/d}, a_1^{\Delta/d}, a_2^{\Delta/d}) \in \mathcal{S}_\Delta$, which is arguably the most technically challenging part. Before getting into it, let us give a proof that $\log \mathcal{S}_d$ is indeed convex.

Lemma 3.4. *For $d \geq 2$, $\log \mathcal{S}_d$ is convex.*

Proof. Let $h(w,x,y)$ be the 'homogenisation' of A_{d+1}, i.e.,

$$h(w,x,y) := (d+1)dw^{d-1}xy + (d+1)w^d(x+y) + w^{d+1}.$$

For $w > 0$, $h(w,x,y) = w^{d+1}A_{d+1}(x/w, y/w)$. Then $(a_0,a_1,a_2) \in \mathcal{S}_d$ if and only if

$$a_0w + a_1x + a_2y \geq h(w,x,y)^{\frac{1}{d+1}} \quad \forall w,x,y \geq 0.$$

Indeed, the 'if' part follows from $h(1,x,y) = A_{d+1}(x,y)$; to see the 'only if' part, note that for $w > 0$,

$$a_0w + a_1x + a_2y = w(a_0 + a_1(x/w) + a_2(y/w)) \geq wA_{d+1}(x/w, y/w)^{\frac{1}{d+1}} = h(w,x,y)^{\frac{1}{d+1}},$$

and the case $w = 0$ is clear since $h(0,x,y) = 0$.

We show that for all $(a_0,a_1,a_2), (b_0,b_1,b_2) \in \mathcal{S}_d$, $(\sqrt{a_0b_0}, \sqrt{a_1b_1}, \sqrt{a_2b_2})$ is in $\mathcal{S}_d$, using this description of $\mathcal{S}_d$ in terms of h. Let $(w_1,x_1,y_1) := (\sqrt{b_0/a_0}w, \sqrt{b_1/a_1}x, \sqrt{b_2/a_2}y)$. Then

$$\sqrt{a_0b_0}w + \sqrt{a_1b_1}x + \sqrt{a_2b_2}y = a_0w_1 + a_1x_1 + a_2y_1 \geq h(w_1,x_1,y_1)^{\frac{1}{d+1}}.$$

By swapping the roles of a_i's and b_i's, let $(w_2, x_2, y_2) := (\sqrt{a_0/b_0}w, \sqrt{a_1/b_1}x, \sqrt{a_2/b_2}y)$. Then

$$\sqrt{a_0b_0}w + \sqrt{a_1b_1}x + \sqrt{a_2b_2}y = b_0w_2 + b_1x_2 + b_2y_2 \geq h(w_2, x_2, y_2)^{\frac{1}{d+1}}.$$

It then remains to show that

$$h(w_1, x_1, y_1)\, h(w_2, x_2, y_2) \geq h(\sqrt{w_1w_2},\ \sqrt{x_1x_2},\ \sqrt{y_1y_2})^2.$$

Let $\mathbf{u}_i := (w_i, x_i, y_i)$ for $i = 1, 2$ and $\mathbf{v} := (\sqrt{w_1w_2}, \sqrt{x_1x_2}, \sqrt{y_1y_2})$. Write $h(w, x, y) = \sum c_{\alpha,\beta,\gamma} w^\alpha x^\beta y^\gamma$, i.e., the monomial representation. As $c_{\alpha,\beta,\gamma} > 0$ always, the Cauchy–Schwarz inequality gives

$$\begin{aligned} h(\mathbf{v}) &= \sum c_{\alpha,\beta,\gamma}(w_1w_2)^{\alpha/2}(x_1x_2)^{\beta/2}(y_1y_2)^{\gamma/2} = \sum \sqrt{c_{\alpha,\beta,\gamma}w_1^\alpha x_1^\beta y_1^\gamma}\sqrt{c_{\alpha,\beta,\gamma}w_2^\alpha x_2^\beta y_2^\gamma} \\ &\leq \Big(\sum c_{\alpha,\beta,\gamma}w_1^\alpha x_1^\beta y_1^\gamma\Big)^{1/2}\Big(\sum c_{\alpha,\beta,\gamma}w_2^\alpha x_2^\beta y_2^\gamma\Big)^{1/2} = h(\mathbf{u}_1)^{1/2}h(\mathbf{u}_2)^{1/2}. \end{aligned}$$

□

3.2 Proof of Lemma 3.3

To sketch our proof of Lemma 3.3, first recall that Lemma 3.1 showed that $A_{d+1}(x, y)^{\frac{1}{d+1}}$ is concave on $(\mathbb{R}_{\geq 0})^2$, so its tangent plane always lies above the surface $z = A_{d+1}(x, y)^{\frac{1}{d+1}}$. In other words, the boundary of $\mathcal{S}_d$ consists of those $(a_0, a_1, a_2) \in (\mathbb{R}_{\geq 0})^3$ such that $z = a_0 + a_1x + a_2y$ is tangent to $z = A_{d+1}(x, y)^{\frac{1}{d+1}}$ at a point in the nonnegative octant. Thus, to show whether a point (v_0, v_1, v_2) is in $\mathcal{S}_\Delta$, it is convenient to compute (or at least estimate) the value v_0^*, where (v_0^*, v_1, v_2) is in the boundary of $\mathcal{S}_\Delta$, and check if $v_0 \geq v_0^*$. It turns out that v_0^* essentially depends on the product v_1v_2; see Lemma 3.5 for the precise statement. This allows us to reduce Lemma 3.3 to the case when $\lambda = \mu$. After the reduction, proving a still technical but low-dimensional inequality concludes the proof.

For $a_1, a_2 > 0$, let

$$\Phi_\Delta(a_1, a_2) := \inf\{a_0 \geq 0 : (a_0, a_1, a_2) \in \mathcal{S}_\Delta\} = \sup_{x,y\geq 0}\big(A_{\Delta+1}(x, y)^{\frac{1}{\Delta+1}} - a_1x - a_2y\big). \tag{3.7}$$

Then for a given $(w_0, w_1, w_2) \in (\mathbb{R}_{>0})^3$, $(w_0, w_1, w_2) \in \mathcal{S}_\Delta$ if and only if $w_0 \geq \Phi_\Delta(w_1, w_2)$. The following lemma provides an upper bound of $\Phi_\Delta(a_1, a_2)$ which only depends on a_1a_2 up to a simple term.

Lemma 3.5. *There is a function $\Psi_\Delta\colon (0, 1) \to \mathbb{R}$ such that for all $a_1, a_2 > 0$ with $a_1a_2 < 1$,*

$$\Phi_\Delta(a_1, a_2) \leq \Psi_\Delta(a_1a_2) + (a_1 + a_2)/\Delta.$$

Moreover, if $a_1 = a_2$, then equality holds.

Proof. We want to identify the point $(x_*, y_*) \in \mathbb{R}^2$ at which

$$\partial_x\big(A_{\Delta+1}(x, y)^{\frac{1}{\Delta+1}}\big) = a_1 \quad \text{and} \quad \partial_y\big(A_{\Delta+1}(x, y)^{\frac{1}{\Delta+1}}\big) = a_2.$$

Denote by $\xi := A_{\Delta+1}(x_*, y_*)^{\frac{1}{\Delta+1}}$. Since

$$\begin{aligned} \partial_x\big(A_{\Delta+1}(x, y)^{\frac{1}{\Delta+1}}\big) &= (\Delta y + 1)\, A_{\Delta+1}(x, y)^{-\Delta/e(\Delta+1)} \quad \text{and} \\ \partial_y\big(A_{\Delta+1}(x, y)^{\frac{1}{\Delta+1}}\big) &= (\Delta x + 1)\, A_{\Delta+1}(x, y)^{-\Delta/(\Delta+1)}, \end{aligned}$$

$a_1 = (1+\Delta y_*)\xi^{-\Delta}$ and $a_2 = (1+\Delta x_*)\xi^{-\Delta}$. Using the identity $(\Delta+1)(\Delta x+1)(\Delta y+1) = \Delta A_{\Delta+1}(x, y)+1$, ξ is a zero of the polynomial $f(X) := (\Delta + 1)a_1a_2X^{2\Delta} - \Delta X^{\Delta+1} - 1$. The following claim, together with the assumption $0 < a_1a_2 < 1$, guarantees the existence of a unique zero.

Claim 3.6. *For $0 < s < 1$, $f(X)$ has a unique zero $\xi_\Delta(s)$ on $(1, \infty)$.*

Proof. Note that $f(0) = -1 < 0$, $f(1) = (\Delta + 1)(s - 1) < 0$, and $f(X) \to \infty$ as $X \to \infty$. In addition, $f'(X) = \Delta(\Delta + 1)X^\Delta(2sX^{\Delta-1} - 1)$ has at most one positive zero $\alpha := (2s)^{-\frac{1}{\Delta-1}}$. Also, $f'(X) < 0$ on $(0, \alpha)$ and $f'(X) > 0$ on (α, ∞). Thus, f has a unique zero on $(0, \infty)$, which must be larger than 1. ■

To emphasise the dependency on a_1a_2, write $A_{\Delta+1}(x_*,y_*)^{\frac{1}{\Delta+1}} = \xi_\Delta(a_1a_2) > 1$. This determines

$$x_* = \frac{a_2\xi_\Delta(a_1a_2)^\Delta - 1}{\Delta} \quad \text{and} \quad y_* = \frac{a_1\xi_\Delta(a_1a_2)^\Delta - 1}{\Delta}. \tag{3.8}$$

In general, this (x_*,y_*) is not in $(\mathbb{R}_{\ge 0})^2$. Instead, we consider a convex set containing $(\mathbb{R}_{\ge 0})^2$ and (x_*,y_*); we claim $\Omega := \{(x,y)\in\mathbb{R}^2 : A_{\Delta+1}(x,y) > 0,\ y > -1/\Delta\}$ is a desired one. That $(\mathbb{R}_{\ge 0})^2 \subseteq \Omega$ is clear, and $A_{\Delta+1}(x_*,y_*) = \xi_\Delta(a_1a_2)^{\Delta+1} > 1$. In addition, $y_* > -1/\Delta$ follows from $\xi_\Delta(a_1a_2) > 1$ and $a_1 > 0$. To see that Ω is convex, let

$$g(y) := -\frac{(\Delta+1)y+1}{(\Delta+1)(\Delta y+1)} \quad \text{so that} \quad \Omega = \{(x,y)\in\mathbb{R}^2 : x > g(y),\ y > -1/\Delta\}.$$

Then $g''(y) = 2\Delta(\Delta+1)^{-1}(\Delta y+1)^{-3} > 0$ on $y > -1/\Delta$, so g is convex on $(-1/\Delta,\infty)$. Hence, Ω is convex. Now by Lemma 3.1, $A_{\Delta+1}(x,y)^{\frac{1}{\Delta+1}}$ is concave on Ω, so

$$\begin{aligned}\Phi_\Delta(a_1,a_2) &\le \sup_{(x,y)\in\Omega}\left(A_{\Delta+1}(x,y)^{\frac{1}{\Delta+1}} - a_1x - a_2y\right) = A_{\Delta+1}(x_*,y_*)^{\frac{1}{\Delta+1}} - a_1x_* - a_2y_* \\ &= \xi_\Delta(a_1a_2) - (2/\Delta)a_1a_2\xi_\Delta(a_1a_2)^\Delta + (a_1+a_2)/\Delta.\end{aligned}$$

Let $\Psi_\Delta(s) := \xi_\Delta(s) - (2/\Delta)s\xi_\Delta(s)^\Delta$ for $0 < s < 1$ so that $\Phi_\Delta(a_1,a_2) \le \Psi_\Delta(a_1a_2) + (a_1+a_2)/\Delta$.

To see the 'moreover' part, suppose $a_1 = a_2 = \sqrt{s}$ for some $s > 0$. Then by definition of ξ_Δ,

$$(\Delta+1)s\xi_\Delta(s)^{2\Delta} = \Delta\xi_\Delta(s)^{\Delta+1} + 1 > \Delta+1,$$

so $a_1\xi_\Delta(a_1a_2)^\Delta = a_2\xi_\Delta(a_1a_2)^\Delta > 1$. Then $x_* = y_* > 0$ by (3.8) and therefore,

$$\Phi_\Delta(a_1,a_2) = A_{\Delta+1}(x_*,y_*)^{\frac{1}{\Delta+1}} - a_1x_* - a_2y_* = \Psi_\Delta(a_1a_2) + (a_1+a_2)/\Delta. \qquad \square$$

The following inequality will allow us to check the condition $a_1a_2 < 1$ in Lemma 3.5 holds in the proof of Lemma 3.3.

Proposition 3.7. *Let $d \ge 1$ and $x,y \ge 0$. Then $(dx+1)^{d+1}(dy+1)^{d+1} \le \big((d+1)dxy+(d+1)(x+y)+1\big)^{2d}$. Equality holds if and only if $x = y = 0$.*

Proof. Let $t := x+y+dxy$. Then the desired inequality is equivalent to $(dt+1)^{d+1} \le ((d+1)t+1)^{2d}$. Let $f(t) := 2d\log((d+1)t+1) - (d+1)\log(dt+1)$. Then $f(0) = 0$ and for $t \ge 0$,

$$f'(t) = d(d+1)\frac{(d-1)t+1}{(dt+1)((d+1)t+1)} > 0. \qquad \square$$

Now we are ready to prove Lemma 3.3.

Proof of Lemma 3.3. For brevity, let

$$A := A_d(\lambda,\mu), \qquad B := B_d(\mu), \qquad C := B_d(\lambda), \qquad D := A_{d+1}(\lambda,\mu),$$

and

$$w_0 := A^{\frac{\Delta}{d}}D^{-\frac{\Delta}{d+1}}, \qquad w_1 := B^{\frac{\Delta}{d}}D^{-\frac{\Delta}{d+1}}, \qquad w_2 := C^{\frac{\Delta}{d}}D^{-\frac{\Delta}{d+1}}. \tag{3.9}$$

Then the goal is to show that

$$(w_0,w_1,w_2) \in \mathcal{S}_\Delta.$$

Note that

$$A = A(B,C) := \frac{(d-1)BC+B+C-1}{d} \quad \text{and} \quad D = \frac{(d+1)BC-1}{d}. \tag{3.10}$$

We have $w_1w_2 \le 1$ since $B^{d+1}C^{d+1} \le D^{2d}$ by Proposition 3.7. If $w_1w_2 = 1$, then equality holds in Proposition 3.7, implying $\lambda = \mu = 0$. Then the goal reduces to check $(1,1,1) \in \mathcal{S}_\Delta$. This indeed holds since for all $x,y \ge 0$,

$$\begin{aligned}(1+x+y)^{\Delta+1} &\ge \binom{\Delta+1}{2}(x+y)^2 + (\Delta+1)(x+y) + 1 \ge (\Delta+1)\Delta xy + (\Delta+1)(x+y) + 1 \\ &= A_{\Delta+1}(x,y).\end{aligned}$$

Reduction to the symmetric case. In what follows, assume $w_1 w_2 < 1$. Our first step is to reduce it to the case $\lambda = \mu$. Recall that $(w_0, w_1, w_2) \in \mathcal{S}_\Delta$ is equivalent to $w_0 \geq \Phi_\Delta(w_1, w_2)$. By Lemma 3.5,

$$w_0 - \Phi_\Delta(w_1, w_2) \geq w_0 - \frac{w_1 + w_2}{\Delta} - \Psi_\Delta(w_1 w_2).$$

Consider w_0, w_1, and w_2 as functions of B and C, which is possible due to (3.9) and (3.10). Note that $w_1 w_2$ is constant when BC is fixed, and hence, so is $\Psi_\Delta(w_1 w_2)$. We claim that if BC is fixed, then $w_0 - (w_1 + w_2)/\Delta$ is minimized when $B = C$. Let $K := (BC)^{1/2}$ and write $B = B(t) := Ke^t$ and $C = C(t) := Ke^{-t}$. Without loss of generality, assume $B \geq C$ so that $t \geq 0$. Let $p := \Delta/d$. Then

$$\frac{d}{dt} A(B, C) = \frac{B'(t) + C'(t)}{d} = \frac{B - C}{d} \quad \text{and} \quad \frac{d}{dt}(B^p + C^p) = p(B^p - C^p).$$

By (3.10), D is fixed and thus, by (3.9),

$$\frac{d}{dt}(w_0 - (w_1 + w_2)/\Delta) = d^{-1}\big(p(B - C)A^{p-1} - (B^p - C^p)\big)D^{-\frac{\Delta}{d+1}}.$$

This is nonnegative, since $A \geq B \geq C$ so that

$$p(B - C)A^{p-1} = \int_C^B pA^{p-1}\, d\zeta \geq \int_C^B p\zeta^{p-1}\, d\zeta = B^p - C^p.$$

Thus, $w_0 - (w_1 + w_2)/\Delta$ is minimized when $t = 0$. Therefore,

$$\begin{aligned} w_0(B, C) - \Phi_\Delta(w_1(B, C), w_2(B, C)) &\geq w_0(B, C) - \frac{w_1(B, C) + w_2(B, C)}{\Delta} - \Psi_\Delta(w_1(B, C) w_2(B, C)) \\ &\geq w_0(K, K) - \frac{w_1(K, K) + w_2(K, K)}{\Delta} - \Psi_\Delta(w_1(B, C) w_2(B, C)) \\ &= w_0(K, K) - \Phi_\Delta(w_1(K, K), w_2(K, K)). \end{aligned}$$

Here the equality follows from the 'moreover' part of Lemma 3.5 with $w_1(K, K) = w_2(K, K)$. In addition, $w_1(B, C) w_2(B, C) = w_1(K, K) w_2(K, K)$ since $w_1 w_2$ depends only on BC and $BC = K^2$. Hence, it suffices to consider the case $B = C$, i.e., $\lambda = \mu$.

Analysis of the symmetric case. Let $A_d(\lambda) := A_d(\lambda, \lambda) = d(d-1)\lambda^2 + 2d\lambda + 1$. Recall that the goal $(w_0, w_1, w_2) \in \mathcal{S}_\Delta$ with $\lambda = \mu$ states that for all integer $1 \leq d \leq \Delta$ and reals $\lambda \geq 0$ and $x, y \geq 0$,

$$\big(A_d(\lambda)^{\frac{\Delta}{d}} + (x + y) B_d(\lambda)^{\frac{\Delta}{d}}\big) A_{d+1}(\lambda)^{-\frac{\Delta}{d+1}} \geq A_{\Delta+1}(x, y)^{\frac{1}{\Delta+1}}.$$

As was shown in (3.6), if $d = \Delta$, then the LHS corresponds to the tangent plane of $z = A_{\Delta+1}(x, y)^{\frac{1}{\Delta+1}}$ at $(x, y) = (\lambda, \lambda)$ lying above the RHS, so the inequality holds. Furthermore, if $x + y$ is fixed, then $A_{\Delta+1}(x, y)$, an increasing function of xy, is maximized when $x = y$. Assuming $x = y$, it suffices to show that for any $x \geq 0$, $\lambda \geq 0$, and $1 \leq d \leq \Delta$,

$$\big(A_d(\lambda)^{\frac{\Delta}{d}} + 2B_d(\lambda)^{\frac{\Delta}{d}} \cdot x\big) A_{d+1}(\lambda)^{-\frac{\Delta}{d+1}} \geq A_{\Delta+1}(x)^{\frac{1}{\Delta+1}},$$

which is shown to be true for $d = \Delta$ and any $x, \lambda \geq 0$. Hence, if for each $\lambda \geq 0$, there exists $\alpha \geq 0$ such that

$$\big(A_d(\lambda)^{\frac{\Delta}{d}} + 2B_d(\lambda)^{\frac{\Delta}{d}} \cdot x\big) A_{d+1}(\lambda)^{-\frac{\Delta}{d+1}} \geq \big(A_\Delta(\alpha) + 2B_\Delta(\alpha) \cdot x\big) A_{\Delta+1}(\alpha)^{-\frac{\Delta}{\Delta+1}}$$

for all $x \geq 0$, then we are done.

To prove $ax + b \geq a'x + b'$ for all $x \geq 0$, it is enough to show $a \geq a'$ and $b \geq b'$. In our case, this means that it is enough to find $\alpha \geq 0$ such that

$$A_d(\lambda)^{\frac{\Delta}{d}} A_{d+1}(\lambda)^{-\frac{\Delta}{d+1}} \geq A_\Delta(\alpha) A_{\Delta+1}(\alpha)^{-\frac{\Delta}{\Delta+1}} \quad \text{and} \quad B_d(\lambda)^{\frac{\Delta}{d}} A_{d+1}(\lambda)^{-\frac{\Delta}{d+1}} \geq B_\Delta(\alpha) A_{\Delta+1}(\alpha)^{-\frac{\Delta}{\Delta+1}}.$$

Rearranging these gives

$$\frac{A_\Delta(\alpha)^{\frac{1}{\Delta}}}{A_{\Delta+1}(\alpha)^{\frac{1}{\Delta+1}}} \leq \frac{A_d(\lambda)^{\frac{1}{d}}}{A_{d+1}(\lambda)^{\frac{1}{d+1}}} \quad \text{and} \quad \frac{B_\Delta(\alpha)^{\frac{1}{\Delta}}}{A_{\Delta+1}(\alpha)^{\frac{1}{\Delta+1}}} \leq \frac{B_d(\lambda)^{\frac{1}{d}}}{A_{d+1}(\lambda)^{\frac{1}{d+1}}}.$$

We may assume $d = \Delta - 1$, since we can use induction on Δ to extend this for smaller values of d. Thus, we want to show that there exists $\alpha = \alpha(\lambda, d) \geq 0$ such that

$$\frac{A_{d+1}(\alpha)^{\frac{1}{d+1}}}{A_{d+2}(\alpha)^{\frac{1}{d+2}}} \leq \frac{A_d(\lambda)^{\frac{1}{d}}}{A_{d+1}(\lambda)^{\frac{1}{d+1}}} \quad \text{and} \quad \frac{B_{d+1}(\alpha)^{\frac{1}{d+1}}}{A_{d+2}(\alpha)^{\frac{1}{d+2}}} \leq \frac{B_d(\lambda)^{\frac{1}{d}}}{A_{d+1}(\lambda)^{\frac{1}{d+1}}}. \tag{3.11}$$

For $k \in \{d, d+1\}$, let $H_k(x) := A_k(x)/B_k(x)$. Then

$$H_k'(x) = \frac{k((k-1)(kx^2+2x)+1)}{B_k(x)^2} > 0, \qquad H_k(0) = 1, \qquad H_k(x) \to \infty \text{ as } x \to \infty, \tag{3.12}$$

so $H_k(x)^{1/k}$ is strictly increasing on $[0, \infty)$ with $H_k(x)^{1/k} \geq 1$. Hence, for each $s \geq 1$ there is a unique $x_k = x_k(s) \geq 0$ such that $H_k(x_k)^{1/k} = s$. Let $s := H_d(\lambda)^{\frac{1}{d}} \geq 1$ and $\alpha := x_{d+1}(s)$. Then $\lambda = x_d(s)$ by definition and $H_{d+1}(\alpha)^{\frac{1}{d+1}} = s = H_d(\lambda)^{\frac{1}{d}}$. If the second condition in (3.11) holds for this $\alpha = \alpha(\lambda, d)$, i.e.,

$$\frac{B_{d+1}(\alpha)^{\frac{1}{d+1}}}{A_{d+2}(\alpha)^{\frac{1}{d+2}}} \leq \frac{B_d(\lambda)^{\frac{1}{d}}}{A_{d+1}(\lambda)^{\frac{1}{d+1}}}, \tag{3.13}$$

then it implies the first condition, as

$$\frac{A_{d+1}(\alpha)^{\frac{1}{d+1}}}{A_{d+2}(\alpha)^{\frac{1}{d+2}}} = \frac{B_{d+1}(\alpha)^{\frac{1}{d+1}}}{A_{d+2}(\alpha)^{\frac{1}{d+2}}} \cdot H_{d+1}(\alpha)^{\frac{1}{d+1}} \leq \frac{B_d(\lambda)^{\frac{1}{d}}}{A_{d+1}(\lambda)^{\frac{1}{d+1}}} \cdot H_d(\lambda)^{\frac{1}{d}} \leq \frac{A_d(\lambda)^{\frac{1}{d}}}{A_{d+1}(\lambda)^{\frac{1}{d+1}}}.$$

For $k \in \{d, d+1\}$, let

$$R_k(s) := \frac{B_k(x_k(s))^{\frac{1}{k}}}{A_{k+1}(x_k(s))^{\frac{1}{k+1}}}.$$

To prove (3.13), it is enough to show $R_{d+1}(s) \leq R_d(s)$ for all $s \geq 1$. To this end, we compute $\partial_s \log R_k(s)$. By writing $x := x_k(s)$ and using $A_k(x)/B_k(x) = s^k$ and $A_{k+1}(x) = A_k(x) + 2xB_k(x) = (s^k + 2x)B_k(x)$,

$$\begin{aligned}\partial_s \log R_k(s) &= \left(\frac{1}{k}\frac{B_k'(x)}{B_k(x)} - \frac{1}{k+1}\frac{A_{k+1}'(x)}{A_{k+1}(x)}\right)\frac{dx}{ds} = \left(\frac{1}{B_k(x)} - \frac{2B_k(x)}{A_{k+1}(x)}\right)\frac{dx}{ds} \\ &= \frac{s^k - 2(k-1)x - 2}{(s^k+2x)B_k(x)}\frac{dx}{ds}.\end{aligned} \tag{3.14}$$

Differentiating both sides of $A_k(x) = s^k B_k(x)$ by s gives $A_k'(x)\frac{dx}{ds} = ks^{k-1}B_k(x) + s^k B_k'(x)\frac{dx}{ds}$, so

$$\frac{dx}{ds} = \frac{ks^{k-1}B_k(x)}{A_k'(x) - s^k B_k'(x)} = \frac{s^{k-1}B_k(x)}{2(k-1)x + 2 - s^k}.$$

This together with (3.14) gives

$$\partial_s \log R_k(s) = -\frac{s^{k-1}}{s^k + 2x_k(s)}. \tag{3.15}$$

Suppose $x_{d+1}(s) \leq s x_d(s)$. Then (3.15) implies

$$\partial_s \log R_{d+1}(s) \leq \partial_s \log R_d(s),$$

and using $R_{d+1}(1) = R_d(1) = 1$, we have $R_{d+1}(s) \leq R_d(s)$. Thus, it suffices to show $x_{d+1}(s) \leq s x_d(s)$.

By (3.12), H_{d+1} is strictly increasing, so $x_{d+1}(s) \leq s x_d(s)$ is equivalent to

$$s^{d+1} = H_{d+1}(x_{d+1}(s)) \leq H_{d+1}(s\, x_d(s)).$$

Write $x := x_d(s)$. By definition,

$$H_{d+1}(sx) = \frac{A_{d+1}(sx)}{B_{d+1}(sx)} = 1 + dsx + \frac{sx}{(d+1)sx + 1}.$$

On the other hand,

$$s^{d+1} = s \cdot s^d = sH_d(x) = s\Big(1 + (d-1)x + \frac{x}{dx+1}\Big).$$

Recall that $s \geq 1$, which implies $d-(d+1)s \leq 0$. Using this, we have

$$\begin{aligned} H_{d+1}(sx) - s^{d+1} &= (1-s) + sx + sx\Big(\frac{1}{(d+1)sx+1} - \frac{1}{dx+1}\Big) \\ &= (1-s) + sx + sx\frac{(d-(d+1)s)x}{((d+1)sx+1)(dx+1)} \\ &\geq (1-s) + sx + sx\frac{(d-(d+1)s)x}{((d+1)s+d)x+1}. \end{aligned} \tag{3.16}$$

Let $Q(x)$ be the expression obtained by multiplying (3.16) by $((d+1)s+d)x+1$. Then

$$Q(x) = (1-s)\big(((d+1)s+d)x+1\big) + sx(2dx+1) = 2dsx^2 + (2s+d-s^2d-s^2)x - (s-1).$$

As $2ds > 0$, $Q(0) \leq 0$, and $Q(s-1) = (d-1)(s-1)^3 \geq 0$, this quadratic polynomial $Q(x)$ is nonnegative on $x \geq s-1$. Indeed, $x = x_d(s) \geq s-1$, as

$$1 + d(s-1) \leq s^d = H_d(x) = 1 + (d-1)x + \frac{x}{dx+1} \leq 1 + dx.$$

Hence, $H_{d+1}(sx) \geq s^{d+1}$, which concludes the proof. □

Remark. While *Aletheia*'s output was closer to an incomplete proof sketch than a complete proof, it contained several significant ideas. For instance, it suggested using the dual set $\mathcal{S}_d$ and its log-convexity; it also provided a preliminary form of Lemma 3.5 and the reduction technique for Lemma 3.3. Based on these insights, we formalised the arguments and filled in the necessary details. In particular, for Lemma 3.4 and the symmetric case of Lemma 3.3, we retained only the statements provided by the model and wrote the proofs entirely independently, using different methods to ensure clarity and simplicity.

4 More towards Conjecture 1.3

Recall that Conjecture 1.3 states that for any graph G and antiferromagnetic edge-weighted graph H,

$$\hom(G,H) \geq \prod_{v \in V(G)} \hom(K_{d_v+1}, H)^{\frac{1}{d_v+1}}. \tag{4.1}$$

We show that this holds if $\Delta(G) \leq 2$.

Theorem 4.1. *Conjecture 1.3 is true if* $\Delta(G) \leq 2$.

Proof. Since both sides of (4.1) are multiplicative over the connected components of G, it suffices to consider the case when G is connected. Thus, we assume G is either a path or a cycle. Let $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n$ be the spectrum of H.

Suppose first that G is an ℓ-edge path P_ℓ where $\ell \geq 1$. By a classical result of Cvetković, Doob, and Sachs [CDS80, Theorem 1.10] (also see [CRS10, Theorem 1.3.5] for a recent source), there exist nonnegative reals $p_1, \ldots, p_n$, $\sum_{i=1}^n p_i = n$, such that the (weighted) number of m-edge walks in H is $p_1\mu_1^m + \cdots + p_n\mu_n^m$ for each m. We then have, by using the antiferromagnetism condition $\mu_i \leq 0$ and $|\mu_i| \leq \mu_1$ for each $i > 1$,

$$\hom(P_\ell, H) = p_1\mu_1^\ell + \cdots + p_n\mu_n^\ell \geq \mu_1^{\ell-1}(p_1\mu_1 + \cdots + p_n\mu_n).$$

Again by the antiferromagnetism, $\mu_1^3 \geq \mu_1^3 + \cdots + \mu_n^3$, so

$$\begin{aligned} \mu_1^{\ell-1}(p_1\mu_1 + \cdots + p_n\mu_n) &\geq (\mu_1^3 + \cdots + \mu_n^3)^{\frac{\ell-1}{3}}(p_1\mu_1 + \cdots + p_n\mu_n) \\ &= \hom(K_3, H)^{\frac{\ell-1}{3}} \hom(K_2, H), \end{aligned}$$

which is the desired bound.

Suppose now that G is a copy of C_ℓ, the cycle of length $\ell \geq 3$. Then, by using $|\mu_i| \leq \mu_1$ for each i,

$$\hom(C_\ell, H) = \mu_1^\ell + \cdots + \mu_n^\ell \geq \mu_1^{\ell-3}(\mu_1^3 + \cdots + \mu_n^3) \geq (\mu_1^3 + \cdots + \mu_n^3)^{\frac{\ell}{3}} = \hom(K_3, H)^{\frac{\ell}{3}},$$

where the last inequality uses $\mu_1^3 \geq \mu_1^3 + \cdots + \mu_n^3$. This concludes the proof. □

We say that an edge-weighted graph H is a *q-spin antiferromagnetic model* if it is antiferromagnetic and $|V(H)| = q$. For instance, $H =$ and any graph obtained by assigning positive edge weights to the two edges of H are 2-spin antiferromagnetic models. Theorem 1.2 hence shows that Conjecture 1.3 holds for all 2-spin antiferromagnetic models of such kind. Let K_{q+1}° be the graph obtained from K_{q+1} by attaching a loop to one vertex, which is a $(q+1)$-spin antiferromagnetic model. This is one of the most important $(q+1)$-spin antiferromagnetic models, as the support of an antiferromagnetic graph is always a blow-up of K_{q+1} or K_{q+1}° after removing isolated vertices [LOS25, Theorem 3.2]. Our goal is to obtain a slightly weaker bound than (4.1) for these graphs.

As was described in the introduction, when the host graph H is $K_2^{\circ} =$, a homomorphism from G to H corresponds to an independent set of G. When $H = K_{q+1}^{\circ}$, a homomorphism from G to H corresponds to a *semiproper colouring* of G, a relaxation of proper vertex colourings by allowing *looped* colours that can be used with no constraints. In particular, a homomorphism from G to K_{q+1}° corresponds to a semiproper colouring of G with q proper colours and one 'free' colour.

Let q be a positive integer. For reals $\lambda^{(i)}$, $1 \le i \le q$, define

$$Z_G^{(q)}(\lambda^{(1)}, \dots, \lambda^{(q)}) := \sum_{(I_1,\dots,I_q)\in\mathcal{I}^{(q)}(G)} \prod_{i=1}^{q} (\lambda^{(i)})^{|I_i|},$$

where $\mathcal{I}^{(q)}(G)$ consists of all q-tuples $(I_1, \dots, I_q)$ of pairwise disjoint (possibly empty) independent sets of G. In particular, $Z_G^{(q)}(1, 1, \dots, 1) = \hom(G, K_{q+1}^{\circ})$. The multiaffine variant of $Z_G^{(q)}$ is defined as follows: for tuples of reals $\boldsymbol{\lambda}^{(i)} = (\lambda_v^{(i)})_{v\in V(G)}$, $1 \le i \le q$,

$$Z_G^{(q)}(\boldsymbol{\lambda}^{(1)}, \dots, \boldsymbol{\lambda}^{(q)}) := \sum_{(I_1,\dots,I_q)\in\mathcal{I}^{(q)}(G)} \prod_{i=1}^{q} \prod_{v\in I_i} \lambda_v^{(i)}.$$

If $q = 1$, $Z_G^{(1)}(\boldsymbol{\lambda}^{(1)})$ is exactly the multivariate independence polynomial $Z_G(\boldsymbol{\lambda}^{(1)})$.

We conjecture that the following analogous statement of Theorem 1.1 holds for $Z_G^{(q)}$. This is a vertex-weighted version of Conjecture 1.3 when $H = K_{q+1}^{\circ}$.

Conjecture 4.2. *Let G be a graph and $\boldsymbol{\lambda}^{(i)} = (\lambda_v^{(i)})_{v\in V(G)}$, $1 \le i \le q$ be tuples of nonnegative reals. Then*

$$Z_G^{(q)}(\boldsymbol{\lambda}^{(1)}, \dots, \boldsymbol{\lambda}^{(q)}) \ge \prod_{v\in V(G)} Z_{K_{d_v+1}}^{(q)}(\lambda_v^{(1)}, \dots, \lambda_v^{(q)})^{\frac{1}{d_v+1}}.$$

We shall prove a weaker bound as an application of Theorem 1.1, which matches the bound in the conjecture in an asymptotic sense. For graphs G and K, their *Cartesian product* (or *box product*) $G \,\square\, K$ is the graph on $V(G) \times V(K)$ in which two vertices (u_1, v_1) and (u_2, v_2) are adjacent if and only if either (i) $u_1 = u_2$ and $v_1v_2 \in E(K)$ or (ii) $u_1u_2 \in E(G)$ and $v_1 = v_2$. Let K_t be a complete graph on $[t] := \{1, \dots, t\}$ so that $V(G \,\square\, K_t) = V(G) \times [t]$. Then the map

$$(I_1, I_2) \mapsto \{(v, 1) : v \in I_1\} \sqcup \{(v, 2) : v \in I_2\}.$$

is a bijection from $\mathcal{I}^{(2)}(G)$ to $\mathcal{I}(G \,\square\, K_2)$. Let us illustrate how $Z_G^{(2)}(\boldsymbol{\lambda}^{(1)}, \boldsymbol{\lambda}^{(2)})$ corresponds to $Z_{G\square K_2}$ with suitable parameters. For simplicity, let $\boldsymbol{\lambda}^{(i)} = (\lambda^{(i)}, \dots, \lambda^{(i)})$, $i = 1, 2$, for $\lambda^{(1)}, \lambda^{(2)} \in \mathbb{R}_{\ge 0}$. Then setting $\boldsymbol{\mu} = (\mu_{(v,i)})_{(v,i)\in V(G\square K_2)}$ where $\mu_{(v,i)} := \lambda^{(i)} \in \mathbb{R}$ for $(v, i) \in V(G \,\square\, K_2)$ gives

$$\begin{aligned} Z_G^{(2)}(\boldsymbol{\lambda}^{(1)}, \boldsymbol{\lambda}^{(2)}) &= \sum_{(I_1,I_2)\in\mathcal{I}^{(2)}(G)} (\lambda^{(1)})^{|I_1|} (\lambda^{(2)})^{|I_2|} = \sum_{I\in\mathcal{I}(G\square K_2)} \prod_{(v,i)\in I} \mu_{(v,i)} \\ &= Z_{G\square K_2}(\boldsymbol{\mu}). \end{aligned}$$

By Theorem 1.1 together with the fact $\deg_{G\square K_2}(v, i) = d_v + 1$ where $d_v := \deg_G(v)$,

$$\begin{aligned} Z_{G\square K_2}(\boldsymbol{\mu}) &\ge \prod_{(v,i)\in V(G\square K_2)} Z_{K_{d_v+2}}(\mu_{(v,i)})^{\frac{1}{d_v+2}} \\ &= \prod_{v\in V(G)} (1 + (d_v + 2)\lambda^{(1)})^{\frac{1}{d_v+2}} (1 + (d_v + 2)\lambda^{(2)})^{\frac{1}{d_v+2}}. \end{aligned}$$

If $\lambda^{(1)}, \lambda^{(2)} = o(1/\Delta(G))$, the factors in this lower bound of $Z_G^{(2)}(\lambda^{(1)}, \lambda^{(2)})$ match the conjectured factors of Conjecture 4.2 up to the linear terms. Indeed, as $\Delta(G) \to \infty$,

$$\begin{aligned}(1+(d_v+2)\lambda^{(1)})^{\frac{1}{d_v+2}}(1+(d_v+2)\lambda^{(2)})^{\frac{1}{d_v+2}} &= \left(1+(d_v+2)(\lambda^{(1)}+\lambda^{(2)})+(d_v+2)^2\lambda^{(1)}\lambda^{(2)}\right)^{\frac{1}{d_v+2}} \\ &= 1+(\lambda^{(1)}+\lambda^{(2)})+o(\lambda^{(1)}+\lambda^{(2)}),\end{aligned}$$

while

$$\begin{aligned}Z^{(2)}_{K_{d_v+1}}(\lambda^{(1)},\lambda^{(2)})^{\frac{1}{d_v+1}} &= (1+(d_v+1)(\lambda^{(1)}+\lambda^{(2)})+(d_v+1)d_v\lambda^{(1)}\lambda^{(2)})^{\frac{1}{d_v+1}} \\ &= 1+(\lambda^{(1)}+\lambda^{(2)})+o(\lambda^{(1)}+\lambda^{(2)}).\end{aligned}$$

If we assume further that $\lambda^{(1)} = \lambda^{(2)} = \lambda = o(1/\Delta(G))$, then

$$\begin{aligned}&(1+(d_v+2)\lambda^{(1)})^{\frac{1}{d_v+2}}(1+(d_v+2)\lambda^{(2)})^{\frac{1}{d_v+2}} \\ &\qquad = \left(1+2(d_v+2)\lambda+(d_v+2)^2\lambda^2\right)^{\frac{1}{d_v+2}} \\ &\qquad = 1+(2\lambda+(d_v+2)\lambda^2)-\frac{1}{2}\frac{1}{d_v+2}\frac{d_v+1}{d_v+2}(2(d_v+2)\lambda)^2+o(\lambda^2) \\ &\qquad = 1+2\lambda-d_v\lambda^2+o(\lambda^2),\end{aligned}$$

while

$$\begin{aligned}Z^{(2)}_{K_{d_v+1}}(\lambda^{(1)},\lambda^{(2)})^{\frac{1}{d_v+1}} &= (1+2(d_v+1)\lambda+(d_v+1)d_v\lambda^2)^{\frac{1}{d_v+1}} \\ &= 1+(2\lambda+d_v\lambda^2)-\frac{1}{2}\frac{1}{d_v+1}\frac{d_v}{d_v+1}(2(d_v+1)\lambda)^2+o(\lambda^2) \\ &= 1+2\lambda-d_v\lambda^2+o(\lambda^2),\end{aligned}$$

so the factors match up to the quadratic terms. This argument generalises to the following theorem, which can be seen as an approximate version of Conjecture 4.2.

Theorem 4.3. *Let G be a graph, $q \geq 1$ be an integer, and $\boldsymbol{\lambda}^{(i)} = (\lambda_v^{(i)})_{v\in V(G)}$, $1 \leq i \leq q$ be tuples of nonnegative reals. Then*

$$Z_G^{(q)}(\boldsymbol{\lambda}^{(1)},\dots,\boldsymbol{\lambda}^{(q)}) \geq \prod_{v\in V(G)}\prod_{i=1}^{q} Z_{K_{d_v+q}}(\lambda_v^{(i)})^{1/(d_v+q)}.$$

In particular, if $\lambda_v^{(i)} = o(1/\Delta(G))$ for all $v \in V(G)$ and $1 \leq i \leq q$, as $\Delta(G) \to \infty$,

$$Z_G^{(q)}(\boldsymbol{\lambda}^{(1)},\dots,\boldsymbol{\lambda}^{(q)}) \geq \prod_{v\in V(G)}\left(Z^{(q)}_{K_{d_v+1}}(\lambda_v^{(1)},\dots,\lambda_v^{(q)})^{\frac{1}{d_v+1}}+o\left(\sum_{i=1}^{q}\lambda_v^{(i)}\right)\right).$$

If in addition, for each $v \in V(G)$, $\lambda_v^{(1)} = \cdots = \lambda_v^{(q)} =: \lambda_v$ for all $1 \leq i \leq q$,

$$Z_G^{(q)}(\boldsymbol{\lambda}^{(1)},\dots,\boldsymbol{\lambda}^{(q)}) \geq \prod_{v\in V(G)}\left(Z^{(q)}_{K_{d_v+1}}(\lambda_v,\dots,\lambda_v)^{\frac{1}{d_v+1}}+o((q\lambda_v)^2)\right).$$

In proving Theorem 4.3, the most technical part is to extend the aforementioned bijection from $\mathcal{I}^{(2)}(G)$ to $\mathcal{I}(G \,\square\, K_2)$. Once we have the bijection, it is straightforward to deduce Theorem 4.3, which we omit.

Lemma 4.4. *Let G be a graph, $q \geq 1$ be an integer, and $\boldsymbol{\lambda}^{(i)} = (\lambda_v^{(i)})_{v\in V(G)}$, $1 \leq i \leq q$ be tuples of nonnegative reals. Let $\boldsymbol{\lambda} := (\lambda_v^{(i)})_{(v,i)\in V(G\square K_q)}$. Then*

$$Z_G^{(q)}(\boldsymbol{\lambda}^{(1)},\dots,\boldsymbol{\lambda}^{(q)}) = Z_{G\square K_q}(\boldsymbol{\lambda}).$$

This may be seen as a vertex-weighted version of a particular case of [KLL16, Lemma 3.1], which is on $\hom(G \,\square\, K, H)$ for graphs G, K, and H. It is clear that Theorem 1.1 and Lemma 4.4 imply the first conclusion of Theorem 4.3, since each vertex (v, i) in $G \,\square\, K_q$ has degree $\deg_G(v) + q - 1$.

Proof of Lemma 4.4. For each independent set I of $G \,\square\, K_q$ and $1 \le i \le q$, let

$$f_i(I) := \{v \in V(G) : (v,i) \in I\}.$$

We claim that $I \mapsto (f_1(I), \dots, f_q(I))$ is a bijection from $\mathcal{I}(G \,\square\, K_q)$ to $\mathcal{I}^{(q)}(G)$. First, if $I \in \mathcal{I}(G \,\square\, K_q)$, then $I \cap (V(G) \times \{i\}) \in \mathcal{I}(G \,\square\, K_q)$ for each i, so $f_i(I) = \{v \in V(G) : (v,i) \in I \cap (V(G) \times \{i\})\} \in \mathcal{I}(G)$. Also, if $f_i(I) \cap f_j(I)$ contains a vertex, say v, for some $i \ne j$, then the edge $(v,i)(v,j)$ lies inside I, a contradiction. Thus, the independent sets $f_i(I)$, $1 \le i \le q$ are mutually disjoint.

On the other hand, suppose $(I_1, \dots, I_q) \in \mathcal{I}^{(q)}(G)$. Let $I := \{(v,i) : v \in I_i, 1 \le i \le q\} \subseteq V(G \,\square\, K_q)$. Then $f_i(I) = I_i$ for each i. Also, $I \in \mathcal{I}(G \,\square\, K_q)$; otherwise, I contains an edge of the form $(u,i)(v,i)$ or $(u,i)(u,j)$ where $u \ne v$ and $i \ne j$, where the former induces an edge uv lying inside I_i, and the latter implies that $I_i \cap I_j$ is nonempty.

Using this bijection,

$$Z_G^{(q)}(\boldsymbol{\lambda}^{(1)}, \dots, \boldsymbol{\lambda}^{(q)}) = \sum_{(I_1,\dots,I_q) \in \mathcal{I}^{(q)}(G)} \prod_{i=1}^{q} \prod_{v \in I_i} \lambda_v^{(i)} = \sum_{I \in \mathcal{I}(G \square K_q)} \prod_{(v,i) \in I} \lambda_v^{(i)} = Z_{G \square K_q}(\boldsymbol{\lambda}). \qquad \square$$

5 Concluding remarks

Conjecture 4.2 for $q > 2$. To generalise Theorem 1.4 to prove Conjecture 4.2 for $q > 2$, one should replace $A_d(x,y)$ by a q-variate function $F_d(x_1, x_2, \dots, x_q) := \sum_{k=0}^{q} (d)_k e_k(\mathbf{x})$, where $(d)_k$ denotes the falling factorial $d(d-1)\cdots(d-k+1)$ and $e_k(\mathbf{x})$ is the elementary symmetric function of degree k. Following the inductive approach, it is natural to replace B_d by the first-order derivatives of F_d. Then $F_d^{1/d}$ is again concave, and the dual set $\mathcal{S}_d$ and its log-convexity generalises with no problems. However, Lemma 3.3, which heavily relies on technical advantages in low dimensions including Lemma 3.5, seems to be the bottleneck. More precisely, as q grows, it gets harder to compute a_0 such that $(a_0, a_1, \dots, a_q)$ is in the boundary of $\mathcal{S}_d$. We believe that $q = 4$ would already require substantially new ideas, whereas it might be possible to handle the case $q = 3$ with analogous techniques to ours.

Extensions to negative fugacity. If G is a (hyper)graph of maximum degree Δ and each λ_v, $v \in V(G)$, is a complex number with modulus at most $\frac{\Delta^\Delta}{(\Delta+1)^{\Delta+1}}$, [GMPST24, Theorem 10] obtains a lower bound of the form

$$Z_G(\boldsymbol{\lambda}) \ge \left(1 - \frac{1}{\Delta+1}\right)^{|V(G)|}.$$

As the RHS is smaller than 1, this is weaker than Theorem 1.1 for nonnegative values of λ_v's. Having seen this comparison, one might wonder whether the range of λ_v's in Theorem 1.1 extends further to negative, or even complex, values. Unfortunately, Theorem 1.1 does not extend to such a range, which was also observed by Gemini 3.0 Deep Think. To elaborate, it is worth noting that substituting the value of λ_0 that makes the partial derivative of $(A+\lambda_0)(1+(\Delta+1)\lambda_0)^{-\frac{1}{\Delta+1}}$ with respect to λ_0 vanish also gives the inequality (2.4). Thus, (2.3) holds if and only if (2.4) holds. But then, analysing the Taylor expansion of $(\Delta+1)A - \Delta B^{\frac{\Delta+1}{\Delta}} - 1$ around $\lambda_v = 0$ proves that (2.4) does not hold locally for some small negative values of λ_v.

Occupancy fractions. When proving inequalities on $Z_G(\lambda)$, or more generally on $\hom(G,H)$, the *occupancy fraction method* is to compare $\frac{d}{d\lambda} \log Z_G(\lambda)$ with $\frac{d}{d\lambda} \log L(\lambda)$, where $L(\lambda)$ is a candidate for the upper/lower bound for $Z_G(\lambda)$. Here $\alpha_G(\lambda) := \frac{\lambda}{|V(G)|} \frac{d}{d\lambda} \log Z_G(\lambda)$, the so-called *occupancy fraction*, has a clear combinatorial interpretation; it corresponds to the expected size of the randomly sampled independent set according to the hard-core model. The occupancy fraction method has seen a lot of success in proving combinatorial inequalities, e.g., [DJPR17; DJPR18]. One of the interesting conjectures in the area is due to Davies and Kang [DK25], stating that the inequality

$$\alpha_G(\lambda) \ge \frac{1}{n} \sum_{i=1}^{n} \alpha_{K_{d_i+1}}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} \frac{\lambda}{1+(d_i+1)\lambda} \tag{5.1}$$

holds for all nonnegative λ and any simple graph G on $[n]$. It is not hard to see that this conjecture implies the Sah–Sawhney–Stoner–Zhao inequality, i.e., Theorem 1.1 with all $\lambda_v = \lambda$. Although the regular

case $d_v = d$ was settled by Davies, Jenssen, Perkins, and Roberts [DJPR17] and Davies, Sandhu, and Tan [DST25, Theorem 1.3] proved the irregular case provided $\lambda \le \frac{3}{(\Delta(G)+1)^2}$, the conjecture remains open.

Having seen the Davies–Kang conjecture, it seems plausible to look for a strengthening of Theorem 1.1 in terms of occupancy fractions. To elaborate, for $v \in V(G)$, let $p_v(\boldsymbol{\lambda}) \coloneqq \lambda_v \frac{\partial}{\partial \lambda_v} \log Z_G(\boldsymbol{\lambda})$ be the *vertex occupancy* or *vertex marginal* at v. This is the probability that v is occupied by an independent set randomly sampled according to the multivariate hard-core model. For $t \in \mathbb{R}$, let

$$\alpha_G(t; \boldsymbol{\lambda}) \coloneqq \frac{t}{|V(G)|} \frac{\partial}{\partial t} \log Z_G(t\boldsymbol{\lambda}) = \frac{t}{|V(G)|} \sum_{v \in V(G)} \lambda_v \frac{\partial}{\partial \lambda_v} \log Z_G(t\boldsymbol{\lambda}) = \frac{1}{|V(G)|} \sum_{v \in V(G)} p_v(t\boldsymbol{\lambda}).$$

If $t = \lambda$ and $\boldsymbol{\lambda} = (1, \dots, 1)$, this is exactly $\alpha_G(\lambda)$. Then Theorem 1.1 follows by showing

$$\alpha_G(t; \boldsymbol{\lambda}) \ge \frac{1}{|V(G)|} \sum_{v \in V(G)} \alpha_{K_{d_v+1}}(t\lambda_v), \qquad t \in \mathbb{R}_{\ge 0},\ \boldsymbol{\lambda} \in (\mathbb{R}_{\ge 0})^{V(G)}, \tag{5.2}$$

multiplying both sides by $|V(G)|/t$, and integrating with respect to t over $(0, 1)$. Putting $t = \lambda$ and $\boldsymbol{\lambda} = (1, \dots, 1)$, we see that (5.1) is a special case of (5.2). Moreover, (5.2) is equivalent to showing that for all $\boldsymbol{\lambda} \in (\mathbb{R}_{\ge 0})^{V(G)}$,

$$\sum_{v \in V(G)} p_v(\boldsymbol{\lambda}) \ge \sum_{v \in V(G)} \alpha_{K_{d_v+1}}(\lambda_v).$$

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